

Operating system and system programming (25CS2104E)

Linux File Descriptor and File I/O Management System

Review-I

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Abstract

The Linux File Descriptor and File I/O Management System is a Linux-based system programming project designed to demonstrate how files are created, opened, read, written, and closed using Linux system calls. The project provides a simple interface for performing basic file operations and understanding how file descriptors are used by the operating system to manage open files. It uses system calls such as `open()`, `read()`, `write()`, `lseek()`, and `close()` to perform file input and output operations. The system will allow users to create or open a file, display its contents, add new data, overwrite existing data, and close the file safely. It will also display important information such as the file descriptor returned by the operating system. The project demonstrates important Operating Systems and System Programming concepts, including file descriptors, file handling, system calls, kernel-level file management, and standard input/output. The expected outcome is a simple and user-friendly Linux application that clearly demonstrates how Linux manages files and file descriptors.

Introduction

Linux provides system calls that allow applications to communicate with the operating system for file and I/O operations.

A file descriptor is a non-negative integer assigned by Linux when a process opens a file or another I/O resource.

Common file descriptors:

- 0 – Standard Input (stdin)
- 1 – Standard Output (stdout)
- 2 – Standard Error (stderr)

Major operations: creating/opening files, reading, writing, changing file offset with lseek(), closing files, and handling errors.

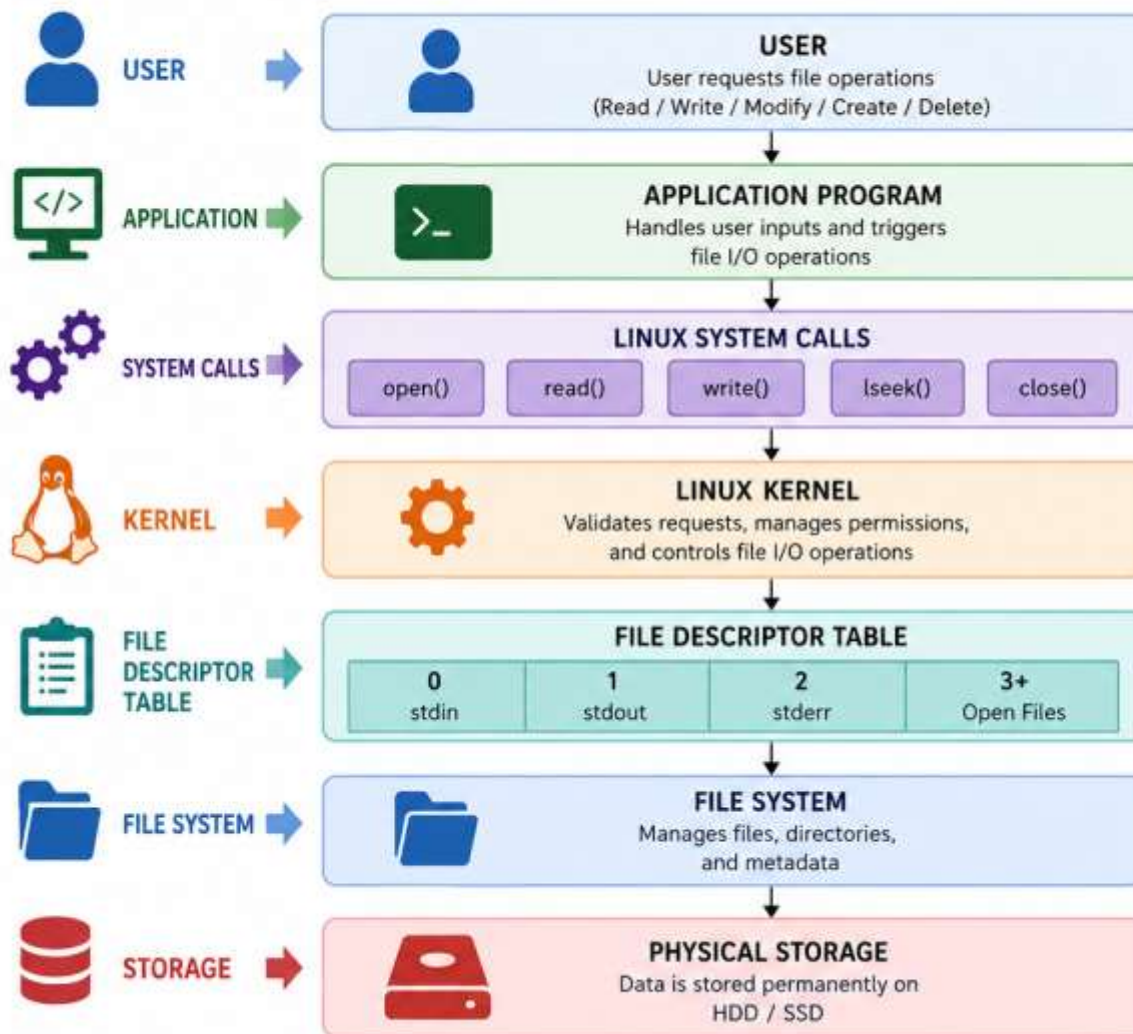
Hardware and Software requirements

List the Requirements in table format

Category	Requirement	Specification / Purpose
Hardware	Processor	Intel/AMD Processor
Hardware	RAM	Minimum 4 GB
Hardware	Storage	Minimum 10 GB free space
Hardware	Keyboard & Mouse	For user input and interaction
Software	Operating System	Ubuntu / Linux
Software	Programming Language	C
Software	Compiler	GCC (GNU Compiler Collection)
Software	System Calls	<code>open()</code> , <code>read()</code> , <code>write()</code> , <code>lseek()</code> , <code>close()</code>
Software	Terminal	Linux Terminal
Software	Code Editor	VS Code / Vim / Nano

SYSTEM ARCHITECTURE

Linux File Descriptor and File I/O Management System



ARCHITECTURE OVERVIEW

The system processes user requests through a sequence of layers. Each layer plays a vital role in ensuring secure and efficient file I/O operations in Linux.



USER

Initiates file operations.



APPLICATION PROGRAM

Takes input and calls system calls.



SYSTEM CALLS

Interface between application and kernel.



LINUX KERNEL

Performs actual I/O operations and access control.



FILE DESCRIPTOR TABLE

Maintains open files and maps them to descriptors.



FILE SYSTEM

Organizes and manages data on storage.



STORAGE

Stores the actual data permanently.

Thank you